

Murmur

<https://d2ajaodgsqz19.cloudfront.net>

Decides which children need an echocardiogram, and says out loud what it missed.

Listening finds one case in ten

Screening schoolchildren in Cambodia and Mozambique, clinical examination found rheumatic heart disease in 2.2 and 2.3 per 1,000. Echocardiography on the same children found 21.5 and 30.4 per 1,000 — roughly ten times as many. The disease affects an estimated 55 million people and claims around 360,000 lives a year, and further damage is prevented with an injected penicillin that has been cheap for decades. The medicine is not what is missing. The echo is.

What it does

Murmur takes a heart-sound recording, cuts it into four-second segments, and runs a small convolutional network over their log-mel spectrograms in the browser. The patient score is the loudest segment, not the average. It returns refer for echocardiography, or no murmur detected.

What it measures

Held-out test patients	174
Test AUC	0.957
Sensitivity at the shipped threshold	94.3%
Specificity	77.0%
Murmurs missed	2 of 35
Children over-referred	32

Split by patient, never by recording. The threshold is chosen for sensitivity, not accuracy: a missed murmur costs a child a valve, a false alarm costs an ultrasound appointment. The two missed murmurs are printed next to the AUC.

What it does not do

- Does not diagnose rheumatic heart disease, or anything else. A murmur is a sound.
- Never deployed or trialled. The screening evidence is about echocardiography, not about this software.
- Not characterised outside its corpus — one programme, one set of equipment.

Sources. WHO rheumatic heart disease fact sheet · Marijon et al., N Engl J Med 2007;357(5):470-6 · CirCor DigiScope, Oliveira et al., IEEE JBHI 2022;26(6):2524-35

Full citations, including figures rejected as unverifiable, in docs/murmur/sources.md